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Changes in work and care trajectories during the transition to motherhood

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A R T I C L E I N F O

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A B S T R A C T

In most mid- and high-income countries, there have been significant demographic, structural, and cultural changes in the past decades. However, we know little about how these changes have shaped women's work patterns during a key life stage: the transition to motherhood. Using longitudinal data from Chile, covering over 30 years of employment histories and three periods of first births (1980–2010), I conduct sequence analysis to identify women's work-care trajectories during an eight-year period of the transition to motherhood. Over time, I find that continuous care work at home has declined, for which education plays a key role, while the chances of working continuously have not changed over time. Instead, I find an increasing trend of unsteady paths that combine paid work with either caretaking or unemployment. I discuss how these changes, as well as their association with education, have important implications for both gender and social inequality.

Women's employment is strongly shaped by family processes, such as the transition to parenthood, where being continuously employed can be particularly challenging (Desai and Waite, 1991; Goldin, 2006; Percheski, 2008). Given the gendered division of labor at home and the social norms and expectations around motherhood, the transition to parenthood is a highly gendered process, mostly affecting women's work (Hays, 1998; Waite et al., 1985; Yavorsky et al., 2015). Even today, 65.1% of mothers of young children in the U.S. participate in the labor force, compared to 94.1% of fathers (BLS, 2018). This gender gap in employment among parents is persistent in many regions outside of the U.S. as well. For instance, in Latin America, 56% of mothers of young children participate in the workforce, compared to 95.2% of men (Gasparini and Marchionni, 2015). Motherhood is an important characteristic that impacts labor market outcomes, both in the short and long term (Abendroth et al., 2014; Correll et al., 2007; Florian, 2018; Looze, 2017).

Motherhood is often analyzed as a discrete event—childless women are compared to mothers, and the effects of being a mother on labor market outcomes is examined. But motherhood can shape women's work plans, workplace behaviors, and experiences in different ways throughout the transition to motherhood, even before getting pregnant (e.g., Bass, 2015; Killewald and Zhuo, 2019; Undurraga, 2018). Therefore, in this paper, I study *work-care trajectories*—that is, sequences of labor force statuses that include paid work, unemployment, and unpaid domestic care work at home—before and after the first birth. Studying trajectories, rather than one point in time estimates of labor force participation (LFP), involves a life course approach of analyzing social pathways that take into account the timing and ordering of events (Elder et al., 2003), which is particularly important during a critical life stage like the transition to motherhood. Higher rates of female LFP at one point in time might not imply more a stable work attachment throughout the life course, which can have significant impacts on future job prospects, wages, and retirement (Arulampalam, 2001; Hotchkiss and

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Pitts, 2007; Madero-Cabib and Fasang, 2016; Pedulla, 2016). Trajectories capture information such as timing and lapses that matter for future outcomes, and that is often missed in cross-sectional measures (Halpern-Manners et al., 2015). An increase in an aggregate cross-sectional measure of mothers' LFP does not tell us whether women are more likely to work throughout the transition to motherhood, if they are moving in and out of work more frequently than before, or if patterns are becoming increasingly heterogeneous.

Important changes related to women's employment have happened in the last decades, such as fertility declines, increases in women's educational attainment, and changes in cultural norms. How have these changes impacted women's work and care-taking patterns during the transition to motherhood? On the one hand, the expansion of education, the decline and postponing of fertility, as well as more egalitarian gender norms (Brewster and Padavic, 2000; Gasparini and Marchionni, 2015; Lesthaeghe, 2010; Percheski, 2008) could indicate that women's employment patterns have become more highly attached to work, increasingly steady and homogenous, similar to those of most men (Stafford et al., 2019; Weisshaar and Cabello-Hutt, 2020 Williams and Han, 2003). On the other hand, strong gendered ideas about parenthood combined with demanding ideal worker expectations, leading to increased levels of work-family conflict (Blair-Loy, 2006; Hennessy, 2009; McDonald, 2000), could indicate that women's employment patterns have become less steady, more complex, and more heterogeneous. In fact, cross-national research on work and family long term patterns has shown that individuals' trajectories have become more individualized, heterogeneous, and destandardized in the past decades (Widmer and Ritschard, 2009; McMunn et al., 2015; Worts et al., 2013). This literature suggests that observing a single cohort of mothers could limit our understanding of change or stability in women's work-care trajectories.

To analyze how the work-care trajectories of women during the transition to motherhood have changed over a period of rapid demographic, structural, and cultural change as well as expand the literature to developing countries, I use the case of Chile, a Latin American mid-to high-income country. Chile, like the United States, is a liberal market-centered state with a high level of income inequality. In contrast with the U.S., while there have been rapid increases in the rates of Chilean women's LFP in the last decades, these are still relatively low—despite the significant declines in fertility and expansion of women's education. While many structural changes are occurring in Chile, traditional gender norms around caretaking remain and are stronger than in the U.S. (Inglehart et al., 2014). Similar to other Latin American countries and some East Asian countries, Chile has strong norms of male-breadwinner and female-caretaker roles (Raymo et al., 2015; Staab, 2012; Undurraga, 2013), that together with women's increased education and employment can lead to particularly high levels of tension in balancing work and family (Anderson and Kohler, 2015; McDonald, 2000). Chile is thus an interesting and important context for studying how motherhood shapes work and care outcomes during rapid demographic transitions.

I use longitudinal and retrospective data from the Survey of Social Welfare, that covers over 30 years of employment histories, to examine the work-care trajectories of women who had their first birth over three periods—the 1980s, 1990s, and 2000s. Using sequence analysis, I examine the work-care trajectories for up to eight years of the transition to motherhood: from two years before to six years after their first childbirth. I then analyze how the likelihood of following different trajectories has changed over time. The findings show that independent of changes in education and fertility, women's work-care trajectories during the transition to motherhood have changed and are becoming increasingly complex and unsteady. Although there has been a decline in continuous care work at home, the chances of being steadily attached to work during the transition to motherhood have remained stable. Further, I show how education shapes the trajectories women follow and how this might impact economic inequality among women. I argue that maternal employment needs to be studied with a trajectory approach because heterogeneity of patterns and interrupted work attachment— often obscured by aggregate trends—is becoming increasingly important. The ability to secure continuous employment during the transition to motherhood can be key in reducing gender inequality and economic inequality among women.

1. A trajectory approach to study employment during the transition to motherhood

The transition to motherhood is an important life event that reshapes work-family arrangements—women may experience the work-family conflict for the first time, and this may shape their levels of attachment to paid work, unemployment, and care work (Waite et al., 1985; Yavorsky et al., 2015). Further, motherhood can shape career aspirations and plans even before childbirth—-women often adapt their work paths in anticipation of future constraints that might come with parenthood (Bass, 2015). Therefore, the way in which becoming a mother shapes work trajectories is a complex process. Fertility scholars have proposed the Theory of Conjunctural Action (TCA) to understand family change as a response to material resources and structure (Johnson-Hanks et al., 2011). At turning points (or conjunctures), such as getting pregnant, individuals have a set of cultural schemas (e.g. ideas or social norms about motherhood) and materials (e.g. parental leave policy or education) available—which are often unevenly distributed—that they draw from to take action, acknowledging both the role of individual agency and social structures (Johnson-Hanks et al., 2011). In sum, work and family arrangements should be interpreted as a social product—as an individual response to available cultural schemas and material structure, such as resources available, policies, norms, and expectations. Research using a trajectory approach has advanced the literature by better documenting the complexity and heterogeneity of these processes of family and work change.

Given this framework that cultural schemas and material resources affect work and family arrangements, it follows that gender will be a key predictor of work patterns over the life course (Moen and Han, 2001). Indeed, we know that across different country contexts, women tend to experience more interrupted work attachment and longer periods out of work than men (e.g., Aisenbrey and Fasang, 2017; Mauro and Yáñez, 2005; Madero-Cabib et al., 2019a,b; Weisshaar and Cabello-Hutt, 2020) leading to more heterogeneity in their work trajectories. Most of past research on women's employment uses cross-sectional measures or aggregate longitudinal measures—but reducing life trajectories to one-point-in-time statuses misses the complexity with which these processes occur at the individual level. A life course approach has the potential of putting social pathways in the center by focusing on *trajectories*—that is, on sequences of roles that are made up of transitions between states (Frytak et al., 2003).

Existing studies that use a trajectory approach to study women's employment show that although most women in the U.S. follow steady work trajectories—around 40% and 58% of all women, depending on the study—a significant proportion of women follow other trajectories that vary in their level of attachment to work across different life stages (Damaske and Frech, 2016; García-Manglano, 2015; Weisshaar and Cabello-Hutt, 2020). Moreover, qualitative research suggests that some women can spend most of their lives in unsteady or interrupted trajectories, moving in and out of jobs and experiencing periods of unemployment (Damaske, 2011). In Chile, where work trajectories have been scarcely studied, recent work shows that about a third of women are continuously employed throughout young adulthood (Madero-Cabib et al., 2019a)—significantly lower than that of the U.S.

In the U.S. context, and focusing on one cohort, some studies have used a trajectory approach to analyze employment trajectories during the transition to motherhood. Focusing on a short period—two months before and one year after the first birth—Lu et al. (2017) find that 57% of new mothers follow a steady work trajectory. However, this is a limited time frame to capture changes occurring due to anticipation to motherhood, pregnancy, as well as longer-term impacts—particularly on work trajectories. A longer-term trajectory approach is especially relevant to the opting out literature since it includes women who may return to work after a long period or have more interrupted trajectories. A recent study addresses this limitation by studying the work trajectories of U.S. women one year before and 18 years after their first birth—they find that around half of the women worked mostly continuously during this period (Killewald and Zhuo, 2019).

Previous work on mothers' employment trajectories often differentiates between employment, unemployment, and out of the labor force—however, this ignores both how unpaid family care work at home is allocated, and the fact that individuals are out of the labor force for multiple reasons (e.g., family responsibilities, education, health issues, among others). Furthermore, we know that not all work lapses have equal impacts; for example, taking time out of work to care for the family has stronger negative effects on hiring prospects than having a lapse because of unemployment (Weisshaar, 2018).

In addition, mothers who are not working and are not actively seeking work—commonly categorized as out of the labor force—might be assumed to be *voluntarily* out of work. However, this notion excludes "discouraged workers," or individuals who give up searching for a job when jobs are not available (Baxandall, 2002; Benati, 2001). Being at home caring for others could be a last resort, a choice constrained by the work conditions individuals see as available to them, by negative past experiences trying to combine work and family, gendered norms and expectations around childrearing, organizational obstacles, and workplace discrimination against mothers (Correll et al., 2007; Damaske, 2011; Johnson-Hanks et al., 2011; Stone, 2007; Undurraga and Barozet, 2015; Williams and Boushey, 2010). In sum, it is important to both measure different types of OLF statuses and to acknowledge the different meanings that these statuses can have.

2. Changes in maternal employment over time

Existing research on maternal employment trajectories has focused on single cohorts, limiting our understandings of change over time. Next, I review demographic, structural, and cultural changes that have occurred in the past decades that I hypothesize have changed the work trajectories mothers follow.

In the past decades, there have been significant increases in women's LFP in most countries in the world, while fertility has declined and family structure has changed (Arriagada, 2004; Goldin, 2006; Lesthaeghe, 2010; Percheski, 2008). Women adapt their employment behavior to their fertility and also their fertility to their employment—although these strategies depend on the context and can vary across different countries (Brewster and Rindfuss, 2000). Nevertheless, mothers are more likely to experience work interruptions or exit the labor market compared with childless women (Budig and England, 2001; Gangl and Ziefle, 2009), and family size is negatively associated with women's employment in developing countries (Cáceres-Delpiano, 2012). Further, fertility timing also shapes the long-term employment trajectories of women over their life course—women who have their first birth later in life are more likely than those who do earlier to follow steady and more attached to work trajectories (Garcia-Manglano, 2015; Sun and Chen, 2017). Thus, other factors being equal, we would expect that as women have fewer children and increasingly have their first birth at older ages, their trajectories would become more continuously attached to work.

The educational gender gap has also narrowed in the past decades, and particularly in Latin America, women have made progress faster than men, even reversing the gap in some countries (Gasparini and Marchionni, 2015). Education plays a key role in mothers' LFP—the human capital theory states that the likelihood of being employed after childbirth is affected by the investment made in human capital (Lundberg and Rose, 2000; Moen and Roehling, 2004). Women who have invested heavily (e.g., have higher levels of education, higher income before childbirth, and more years of work experience) will more likely have a higher attachment to work and will face higher costs of leaving the market (Brewster and Rindfuss, 2000; Desai and Waite, 1991; Moen and Roehling, 2004; Pettit and Hook, 2009). Therefore, as education levels increase, we would expect women to be more continuously attached to the workforce during the transition to motherhood.

Finally, the division of household labor, social norms, and cultural context can impact maternal employment (Boeckmann et al., 2015). In the past decades, there has been a trend towards a decreased stigma around women's work and increased egalitarianism (Brewster and Padavic, 2000; Goldin, 2006; Lesthaeghe, 2010; Percheski, 2008). Gender beliefs and attitudes around parenthood and towards mother's employment can shape women's workforce decisions as well as how women are perceived in the workplace (Correll, 2004; Ridgeway and Correll, 2004; Staab, 2012; Undurraga, 2013). Hence, as these attitudes become more egalitarian, these theories would predict that women will tend to combine work and motherhood earlier and more continuously—this could be reflected in women in younger cohorts being less likely to opt out or to take long periods out of work with the birth of a child, compared to women in older cohorts. At the same time, however, persistent gendered ideals of parenthood may impact the work-family arrangements during the early years of a child, putting women in charge of most of the childcare responsibilities during the transition to parenthood (Blair-Loy, 2006; ComunidadMujer, 2016; Hays, 1998; Hennessy, 2009). An increase in intensive motherhood ideals in a context of competitive and demanding workplaces, as well as rigid and unpredictable schedules, may be pushing women with young children out of the labor force (Blair-Loy, 2006; Williams and Boushey, 2010).

In a context of significant and rapid demographic and cultural change, how has mothers' employment attachment changed? Evidence based on U.S. professional women's employment rates suggests that contrary to the media depiction, there is no evidence of an "opt-out revolution" (Percheski, 2008). Rather, women of younger cohorts in the U.S. seem to be more attached to employment and have shorter employment lapses after the first birth (Macran et al., 1996). Although these findings show that full-time, year-round employment rates have increased in younger cohorts (Percheski, 2008), this does not necessarily mean that individual women's employment trajectories are becoming increasingly stable across the years—and especially during the transition to motherhood.

Existing research has not examined trends of women's work trajectories during the transition to motherhood over time, which is particularly relevant in the context of rapid and significant demographic and structural change. Although not all directly measured in this paper, the changes that have occurred in the past decades inform this analysis as they suggest that mothers' work-care trajectories have changed in important ways.

3. Mothers' labor force participation in Chile

In the past decades, the female labor force participation rate in Chile has significantly increased, but it is still lower than most other Latin American countries, reaching 48.5% in 2017 (INE, 2017). Since the 1960s, fertility has significantly declined, reaching a fertility rate of 1.79 in 2015. Using data from the World Bank and the Chilean census,[1] Fig. 1 shows that fertility declines were followed by female LFP increases, with a lag. As women's integration into the labor market continued to increase, fertility continued to decline. The rapidly increasing trend of women's LFP since the 1990s has recently stalled, and this is not unique to Chile. Since the early 2000s, many Latin American countries have experienced a deceleration in the male/female LFP ratio, and this is particularly pronounced among mothers and lower educated women (Gasparini and Marchionni, 2015).

Low fertility scholarship shows that there is large cross-country heterogeneity in the relationship between fertility and gender equity, and that both institutional and family level gender equity needs to be taken into account to understand this relationship (Anderson and Kohler, 2015; Morgan, 2003; McDonald, 2000). Anderson and Kohler (2015) suggest that many second-wave developing countries experience incoherence between institutional gender equity and family gender equity—as women become increasingly educated and employed in a household context of traditional gender norms where a male-breadwinner/female-caretaker model prevails, like in Chile and other non-high-income countries[2] (Staab, 2012; Undurraga, 2013), they might experience higher levels of work-family conflict together with fertility declines. In Chile, 40% of women are OLF due to permanent family responsibilities (INE, 2017). The work-family conflict could be further exacerbated by the increases in female-headed households and single motherhood (Arriagada, 2004; Blofield and Martínez F., 2014; Esteve et al., 2012).

Women have gained more autonomy in fertility control, but the current stalled, relatively low LFP could be partially explained by incoherence between relatively more gender egalitarian institutional contexts (access to education and work) and persisting male breadwinner/female care-giver norms in Chile. Despite a slow egalitarian trend of more positive attitudes towards women's employment, Chile is very conservative, especially regarding gender norms (Staab, 2012; Undurraga, 2013). In terms of work and family policy, Chile has a predominantly maternalist policy, where care is recognized as a female responsibility without seeking to reduce the gender inequality in the family (Blofield and Martínez Franzoni, 2015). If advances in education and aggregate labor market participation are clashing with traditional gender norms in the family and the workplace, the work-family conflict mothers experience might be increasing, leading to more interrupted and heterogeneous work patterns.

This paper makes important contributions to the literature on maternal employment. First, using a life course framework and an innovative trajectory approach allows me to examine women's employment around motherhood, acknowledging its transitional nature. Second, existing studies often do not distinguish between reasons for being out of the labor force. By using a measure of unpaid care work at home instead of treating all OLF states equally, I use a more specific metric to observe work-family conflict that allows me to capture increased variation among women's work trajectories. Third, existing studies on mothers' work trajectories focus on the U.S. and Western Europe and tend to study one cohort of women.Based on the significant demographic, structural, and cultural transformations that have happened in the past decades, and that closely relate to women's employment, a single cohort approach is limited. This study expands the literature on maternal employment trajectories to middle-income countries that have experienced distinct demographic and women's LFP trends, and for which longitudinal data on work histories is rare. Importantly, by including three cohorts of mothers—that is, women who had their first birth during three periods—this study advances the literature by examining how mothers' work trajectories have changed over time. This is relevant because it is not clear how rapid changes in

[1] I use the Chilean census for the 1960, 1970, and 1982 LFP rates and the World Bank data for all fertility rates (The World Bank, 2018a) and 1990–2015 LFP rates (The World Bank, 2018b).

[2] Wave 6 of the World Value Survey (2011 for the U.S. and 2012 for Chile) shows that the percentage of upper-class women who agree with "When a mother works a child suffers" is much higher in Chile than in the U.S., and in Chile there are also larger class differences: 29% of women (35.5% of men) agree in the lowest three income groups in comparison with a 43.6% of women (51.3% of men) in the highest three income groups. In the U.S. 22.4% of women (37.8% of men) agree in the lowest three income groups in comparison with a 22.8% of women (24.1% of men) in the highest three income groups (Inglehart et al., 2014).

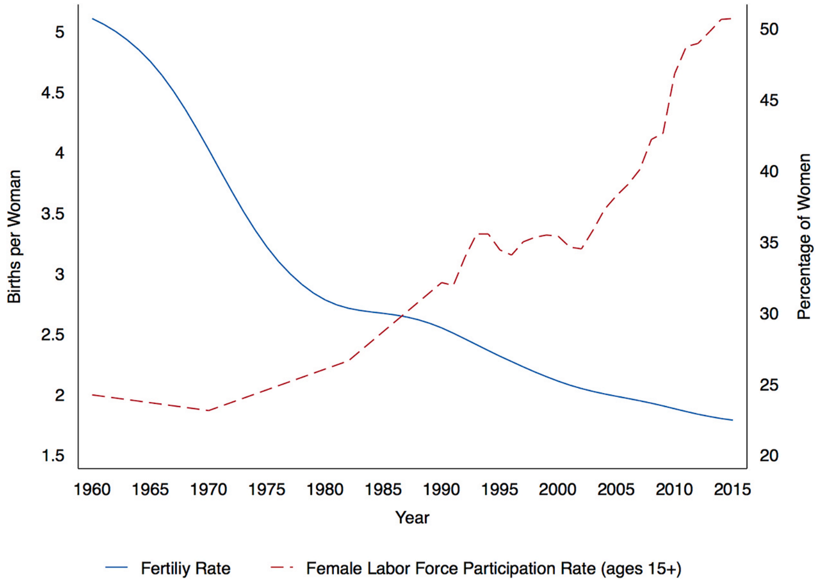

Fig. 1. Female LFP Rate (1990–2015) and Fertility Rate (1960–2015) in Chile. Note: The sources are the Chilean census for the 1960, 1970, and 1982 LFP rates and the World Bank data for all fertility rates (The World Bank, 2018a) and 1990–2015 LFP rates (The World Bank, 2018b).

fertility, education, and gender norms have shaped women's work trajectories during the transition to motherhood. Are the LFP rate increases we see reflecting a trend of women following increasingly steady and attached work trajectories? Or are these changes leading to increased work-family tension and constraints that lead to more interrupted and heterogeneous patterns?

4. Methods

4.1. Data

I use the Survey of Social Welfare (*Encuesta de Protección Social*), conducted by the Ministry of Social Development. This survey is the first long-term panel study in Chile and recognized as Latin America's pioneer survey in collecting longitudinal labor market and social security data. Data have been collected in five waves from 2002 to 2015. By using the retrospective and concurrent work and fertility history data, it is possible to construct monthly work history data for a maximum of thirty-five years, between 1980 and 2015. The survey sample for 2002 is nationally representative of the Chilean population that were registered in the National Retirement System, and the samples from 2004 onwards are nationally representative of the adult population (ages 18 and above).

The analytical sample includes all women who reported having a first child between 1982 (two years after the work histories begin) and 2011.[3] This allows me to analyze a maximum period of two years before and six years after the birth of the first child.

4.2. Identifying the work-care trajectories

To account for the duration and sequence in which work transitions occur, I performed sequence analysis on a sample of women who had a first birth during the period (Abbott and Forrest, 1986; Aisenbrey and Fasang, 2010; Gabadinho et al., 2011; Halpin, 2012).[4] Since most of the changes in mothers' employment have been in workforce participation rather than hours worked, I use labor force status to create the trajectories. Two key variables are used for the sequence analysis: (i) monthly employment status for the period; working, unemployed, or out of the labor force (OLF); and (ii) reason for being OLF, which includes a list of fifteen items that I recoded into care/home (taking care of children, other personal or family responsibilities, pregnancy, housework, and elderly care), and other (i.e. studying, illness, not interested in working).

[3] Examining work trajectories of women during the transition to motherhood, rather than for all women, is key for several reasons. First, the transition to motherhood is an important turning point in the life course, particularly regarding work and family arrangements. Second, the vast majority of Chilean women become mothers at some point in their lives. In 2011, the most recent year I examine, 93% of women age 55 and older had children. Third, the increase in the female LFP since 1990 has been mostly an increase of mothers and married women participating in the labor force (Arriagada, 2004; Larrañaga, 2007), suggesting that mothers' experiences in the workforce are particularly important to understand change over the past decades. Finally, among five South American countries, Chile has the largest motherhood wage penalty, suggesting the importance of motherhood on employment outcomes (Villanueva and Lin, 2019).

[4] I use the R packages TraMineR (Gabadinho et al., 2010, 2011) and WeightedCluster (Studer, 2013) to perform the analysis.

The dataset has 5,187 women who became mothers between 1982 and 2011[5] and have some work history data available. As to missing statuses, I dropped the cases that had the following: an initial gap longer than 12 months, a final gap longer than 24 months, and a continuous internal gap of more than six months (n = 545). For shorter internal gaps, I imputed the monthly labor force status with the previous monthly status available (n = 145 person-months). As a minimum threshold for available data, I set one year pre-birth and four years post-birth—the latter allows me to include women for which a longer post-birth period was not yet observed in the survey. Thus, initial gaps of up to one year and final gaps of up to two years were left as missing (4,039 person-months). Missing statuses at the left side (beginning of the sequence) are included as an additional category, as missing statuses. Missing statuses at the right side (end of the sequence) are voided, indicating that the sequence ended at a different length than other sequences (Gabadinho et al., 2011). Given that my focus is on LFP and unpaid domestic care work, I restricted the sample to women who either were in the labor force (working or unemployed) or were doing unpaid care work at home after childbirth. In other words, I excluded women who studied or were OLF for other reasons post birth (n = 978). The sample size is not sufficiently large to be able to identify the increased diversity of trajectories that are captured when including women who studied or were inactive for other reasons, such as illness, after childbirth. Finally, due to missing data in the variables included in the multinomial models (age at childbirth, education, and parental occupation), an additional 40 cases are dropped. The final sample has 3,624 women and 347,516 person-months. Survey weights were used in all analyses.

Sequence analysis allows us to understand intra-individual variability across time by organizing individual trajectories of states in a chronological way (Blanchard et al., 2014). After formatting the data by the sequences of events or states, sequences are compared by an optimal matching analysis (OMA) that will identify the extent to which individual trajectories are similar, considering the type of status, as well as their timing and order. OMA generates a distance matrix that summarizes individual trajectories' differences and similarities.

Optimal matching distance is the minimal cost needed to transform one sequence into another one. OMA compares how similar or dissimilar the sequences are, based on the number of operations (substitution, insertion, and or deletion) needed to transform one sequence into another, and the total distance is the sum of the costs associated with each needed operation (see Gabadinho et al., 2010, for a detailed example). To do this, substitution costs and insertion-deletion ("indel") costs need to be predefined. In previous research, substitution costs are commonly set as 2 or they are derived from the data—by using the transition rates between two states—while indel costs are commonly set as half of the maximum substitution cost (Aisenbrey and Fasang, 2017; Gabadinho et al., 2010; Macindoe and Abbott, 2004). The optimal distances obtained using these two methods yield very similar results,[6] so I used the cluster solution quality measures as criteria to select the approach that generated the clusters with the better fit. The results presented in the paper are based on the optimal distances calculated with a substitution cost of 2 and an indel cost of 1.

This is followed by hierarchical cluster analysis (Ward) over the OMA distance matrix to identify typologies of sequences.[7] One of the challenges of cluster analysis is to identify the number of clusters or work-care patterns. Several cut-off criteria indices can be calculated to guide this process. I use four measures of the quality of a partition: Average Silhouette Width (weighted) (ASWw), Hubert's Somers' D (HGSD), Point Biserial Correlation (PBC), and Hubert's C (HC).[8] Often, cluster stopping rules suggest different results, so it is important to include both theoretical and pragmatic approaches in selecting the number of clusters to obtain informative classifications (Halpin, 2016:13). Figure A.1 in the Appendix shows the standardized quality measures for up to 10 clusters. The solutions with 5, 6, and 7 clusters are found to have the best quality measures. However, since the next step of this analysis involves analyzing the association between multiple factors and the cluster classification, it is important to avoid having clusters with very small sample sizes, which is what happens when going from 5 clusters to 6 and 7 clusters, as groups that were already relatively small in the 5-cluster solution separate into even smaller groups (see Table A.1 in the Appendix). Considering both the quality measures and the sample size limitation, I selected the 5-cluster solution.[9]

4.3. Predicting membership in work-care trajectories

To analyze change over time as well as the effects of age at first birth, education, and social class background on the likelihood of following different trajectories, I use a multinomial logistic regression. This allows estimating the effect of the period in which the first birth occurred after accounting for differences in fertility timing and educational and class background.

The dependent variable is the work-care trajectory from two years before to six years after the respondent's first birth, as identified by the sequence analysis. The independent variables are: period in which the first birth occurred (P1 = 1982–1989, P2 = 1990–1999, and P3 = 2000–2010) social class background (measured by highest parental occupation), educational level of the mother before

[5] I excluded women who had their first childbirth reported before 14 years old (n = 62).

[6] The results of the analysis using a data-driven approach to set substitution and indel costs are available upon request.

[7] Ward is commonly used in sequence analysis. Another common clustering method (non-hierarchical) that is used in sequence analysis is Partitioning Around Medoids (PAM). Analysis using PAM rather than Ward do no substantively change the results (analysis are available upon request).

[8] ASWw indicates the coherence of the assignments, by calculating between-group distances and within-group homogeneity. HGSD and PBC measures the reproducibility of the distances, and HC measures the gap between the partition obtained and the best theoretically possible partition (for more details see Studer, 2013).

[9] By selecting the 6-cluster solution, the "unsteady work-care" group of the 5-cluster solution would be divided in two groups, while in the 7-cluster solution, in addition to the division of the latter group, the "unsteady work-unemployment" group of the 5-cluster solution would be divided in two groups.

childbirth (1 = Some primary or less; 2 = Some high school, 3 = Some higher education or more), age at childbirth centered at the mean age (22 years old), age at childbirth squared to allow for a non-linear effect of age, and the country level unemployment rates in the year of childbirth.

5. Results

5.1. What are the work-care trajectories mothers follow?

The work-care trajectories of women during the transition to motherhood are heterogeneous, but it is also possible to find some commonalities between them. Fig. 2 presents the trajectories of all women from two years before to six years after their first birth. Each horizontal line represents the sequence followed by one woman—showing a diverse range of sequences.

The cluster analysis of these sequences identifies groups of women who follow relatively similar trajectories. The results suggest five groups of trajectories that are presented in Fig. 3. On the left side of the figure, the index plots show all the individual sequences over time. On the right side of the figure, the modal plots show the modal status of each group over time. The five trajectory groups and their distributions are described in Table 1. A large proportion of women follow either a steady care trajectory (38.92%) or a steady work trajectory (34.42%).

Next, I describe each one of the trajectory groups as well as the characteristics of women who follow them. Table 2 presents a set of characteristics of the five work-care trajectory groups: period and age at childbirth, education, parental occupation, and family and work characteristics during the period. Two measures of sequence characteristics are included: the average number of transitions that occurred in the individual sequences of each trajectory group and the average entropy. The average entropy is a 0-to-1 measure of within group heterogeneity of the states that occur in each time point; where a value of 0 means that all sequences within a group are the same (Gabadinho et al., 2010).

The *steady care* trajectory (38.92%) mostly includes women who spent almost all of the period caring for others at home, even before the birth of their first child (see modal plot in Fig. 3). They have low levels of work and unemployment throughout the period, but as the index plot in Fig. 3 shows, some of the women worked at the beginning of the period and increasingly exited the labor force as they approached childbirth, suggesting a subgroup that "opted out" before childbirth, which is consistent with a 1.24 average number of transitions. Most of these women have some high school education (64.4%) and are from relatively disadvantaged backgrounds. Regarding family formation, their first birth was at 21.59 years old, in average—which is slightly lower than the sample average of 22.54—and a relatively large proportion was married at the time of childbirth (65.7%). This group also has the largest proportion of women who had a second child during the period (65.3%) and the highest average number of children (2.44). Overall, both the average entropy (0.17) and high frequencies in the modal plot (Fig. 3) suggest that this group is relatively homogenous in terms of the sequences they follow.

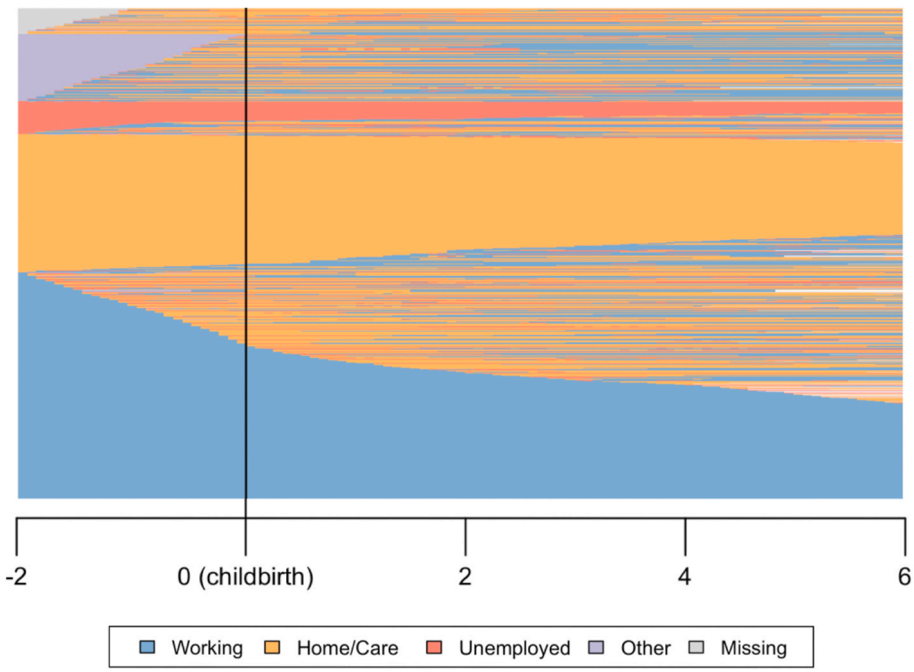

Fig. 2. Work-Care Trajectories during the Transition to Motherhood. Note: Each horizontal line represents the sequence followed by one woman. The white space represents final gaps. This figure should be printed in color (for interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

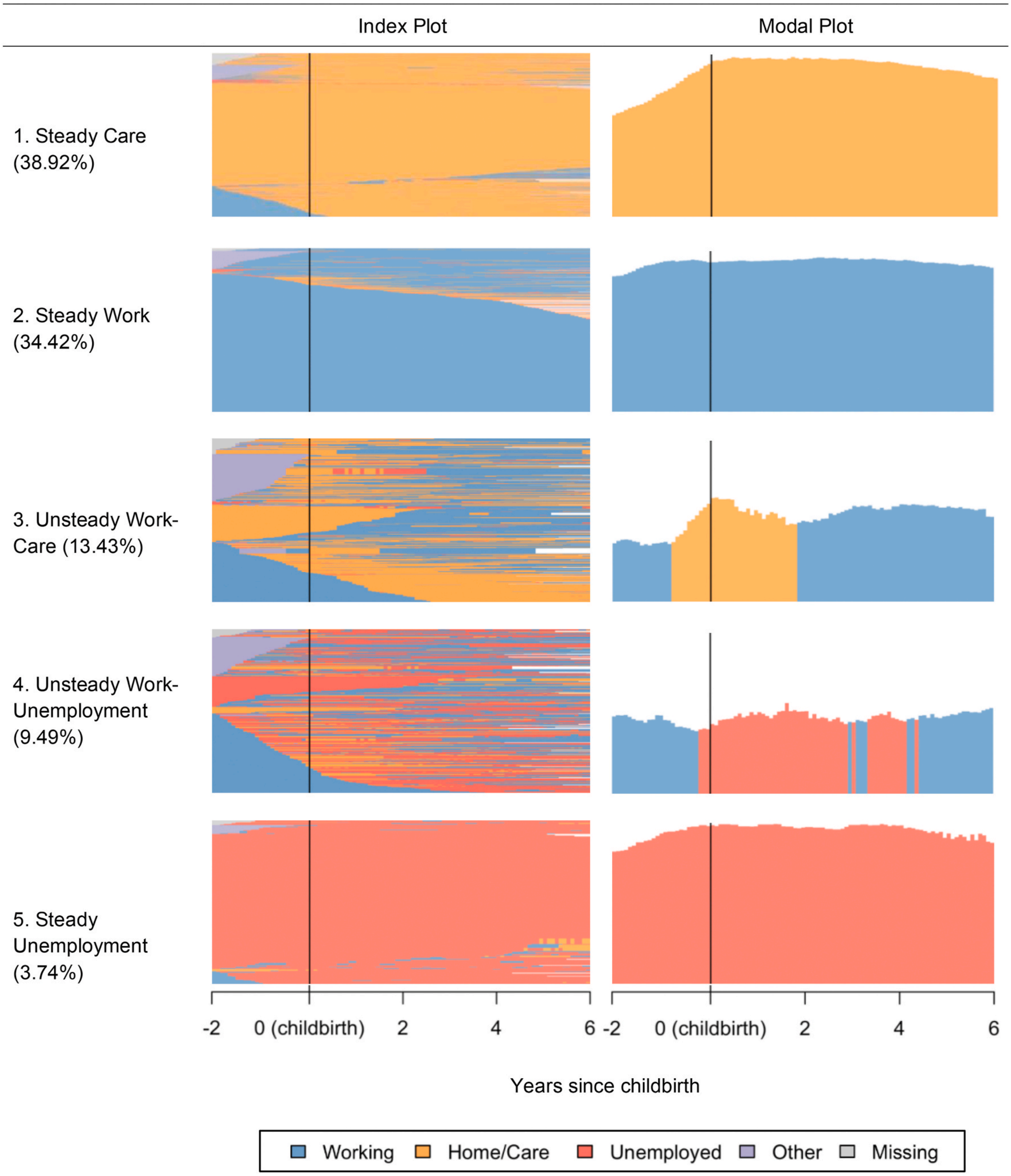

Fig. 3. Work and Care Trajectories by Group. Note: On the index plots, horizontal lines represent the sequence followed by one woman and they are presented as proportions within each group. On the modal plots, vertical bars represent the modal status for each month. This figure should be printed in color (for interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The *steady work* trajectory (34.42%) is a very homogenous group (entropy of 0.11) that includes mostly women who work continuously throughout the transition to motherhood, as the high frequency and steady modal plot shows (Fig. 3). The continuity of work in this trajectory is reflected by a very low average number of transitions (0.95), few and short work lapses, and the fact that they spent the vast majority of the time working, both before and after childbirth. The women in this group are also distinctive as they are from relatively advantaged backgrounds, highly educated, have the highest average age at childbirth (24.59 years old) as well as have fewer children than other groups.

The next two groups are the *unsteady work-care* (13.43%) and the *unsteady work-unemployment* (9.49%). I label these groups as *unsteady* because they have relatively long lapses out of work and the highest average number of transitions among all groups. Both the

Table 1
Description of work-care trajectories.

Group	N	%	Weighted %	Description
1. Steady Care	1,287	35.51	38.92	Most women do care work at home continuously. About a fifth of women work in the pre-birth period and a smaller group works late in the post birth period.
2. Steady Work	1,371	37.83	34.42	Most women work continuously. A small proportion of women study in the pre-birth period.
3. Unsteady Work-Care	482	13.30	13.43	Most women experience a period of care work at home around birth. Before birth, women are at home, working, or studying. In the post-birth period, some women return to work at different time points and work in commonly intermittent and combined with care work at home. Some women "opt out" of work, moving to care work at home.
4. Unsteady Work-Unemployment	361	9.96	9.49	Most women work before birth but experience unemployment after birth. This trajectory combines unemployment, work, and to a lesser extent care work at home, in an intermittent way.
5. Steady Unemployment	123	3.39	3.74	A small proportion of women follow this trajectory of steady unemployment throughout the period, where only a few women work for short periods of time.

Table 2
Descriptive statistics by trajectory group.

	Steady Care	Steady Work	Unsteady Work-Care	Unsteady Work-Unemp.	Steady Unemp.	Total
Period						
1982–1989	50.6%	31.3%	8.9%	4.2%	5.0%	100.0%
	42.4%	*29.7%*	*21.6%*	*14.6%*	*43.5%*	*32.6%*
1990–1999	37.5%	35.7%	13.7%	9.6%	3.6%	100.0%
	40.9%	*44.0%*	*43.4%*	*42.8%*	*40.5%*	*42.4%*
2000–2010	26.1%	36.4%	18.9%	16.2%	2.4%	100.0%
	16.7%	*26.3%*	*35.1%*	*42.7%*	*16.1%*	*24.9%*
Total	38.9%	34.4%	13.4%	9.5%	3.7%	100.0%
	100.0%	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*
Mean Age at childbirth	21.59	24.59	20.86	21.61	22.09	22.54
Education						
Some Primary	32.7%	13.2%	16.9%	21.0%	34.3%	22.8%
Some High School	64.4%	65.4%	75.1%	73.0%	62.8%	67.0%
Some Higher Education	2.9%	21.4%	8.1%	6.1%	3.0%	10.3%
Highest Parental Occupation						
No occupation	12.9%	4.7%	3.7%	4.8%	14.1%	8.1%
Manufacture, agriculture, machine operators, elementary occupations, other	64.3%	56.8%	60.8%	66.4%	67.3%	61.6%
Services	11.6%	13.6%	13.4%	12.9%	9.9%	12.6%
Technical professionals, Clerical support	5.5%	12.7%	7.3%	6.7%	4.8%	8.3%
Managerial/Professional	5.7%	12.2%	14.9%	9.2%	3.9%	9.5%
Family characteristics during the period						
Married at childbirth[a]	65.7%	44.1%	40.3%	46.6%	51.9%	52.6%
Had second child during period	65.3%	41.5%	51.5%	42.0%	56.1%	52.7%
Mean years after had a second child during the period[b]	3.24	3.31	3.58	3.59	3.39	3.34
Total number of children	2.44	1.92	2.00	1.95	2.17	2.15
Work characteristics during the period						
Number of lapses (out of work)	0.44	0.39	1.20	1.85	0.37	0.65
Months working (total)	6.15	89.86	45.32	39.85	2.12	43.27
Lapse duration (months)	4.34	2.96	15.86	23.38	2.81	7.16
Time working pre-birth (%)	13.95	91.55	37.68	44.92	3.60	46.40
Time working post birth (%)	3.89	94.29	50.39	40.38	1.75	44.64
Time unemployed pre-birth (%)	2.04	1.28	2.00	28.93	91.65	7.67
Time unemployed post birth (%)	0.94	1.50	3.09	45.81	98.06	9.31
Sequence characteristics						
Average number of transitions	1.24	0.95	3.30	5.11	1.40	1.24
Average entropy[c]	0.17	0.11	0.51	0.52	0.11	0.17

Note: All percentages and means are weighted. Percentages in italic are column percentages for each work-care trajectory group.

[a] The complete marital histories are not available for 34.9% of the sample mostly because the date of the end of the marriage/cohabitation was not collected in the first wave.

[b] Only among women who had a second child during the period.

[c] Entropy is a measure of within group heterogeneity of the states that occur in each time point (0–1); where a value of 0 means that all sequences within a group are the same.

unsteady work-care and the unsteady work-unemployment trajectories have relatively high within-group heterogeneity, as the level of entropy shows (0.51 and 0.52, respectively). This suggests that compared to other groups, these are more complex. However, the trajectories within each group still share common characteristics and clearly differentiate from the other trajectory groups identified.

In the *unsteady work-care* trajectory, unpaid care work at home predominates around childbirth, but by the second year post-birth, most of these women are in the labor force at some point (see modal plot in Fig. 3). As the index plot shows, there is significant variation in the sequence and timing in which care and work are combined. The trajectories in this group are also intermittent, as the average number of transitions is the second highest among all groups (3.3). Before childbirth, they work 37.68% of the time, on average—after childbirth, this increases to 50.39%. Unemployment slightly increases too, but it is still low overall. Regarding class background, this group is relatively advantaged, but they have slightly below average levels of higher education. The latter could be related with the low average age at childbirth (20.86 years old) as well as the low proportion of marriage at childbirth—suggesting they are more likely to have had a teenage pregnancy.

The *unsteady work-unemployment* trajectory (9.49%) is characterized by high levels of unemployment and the highest number of average lapses out of work, suggesting that this is a negatively unstable trajectory. As the modal plot (Fig. 3) shows, the periods of unemployment become predominant a couple of months before birth and continue to be the mode until around four years after birth. The levels of work decline after childbirth and the levels of unemployment significantly increase—from 28.93% of the pre-birth period to 45.81% of the post-birth period. Further, this is the group with the most heterogenous and interrupted trajectories, as the high average entropy and number of transitions (5.11) show. Compared to the unsteady work-care trajectory, this group has a more disadvantaged class background and lower levels of education.

Finally, the *steady unemployment* trajectory represents a small percentage of the sample (3.74%) that is mostly unemployed throughout the period, as it is confirmed by the high proportion of time spent unemployed as well as the low average number of work lapses. This is also a relatively homogeneous group in terms of the sequences followed, as indicated by a low average entropy (0.11). Relative to all other groups, this group has the most disadvantaged class background as well as the lowest levels of education, suggesting this is a highly vulnerable group of women who experience difficulties getting a job throughout the transition to motherhood.

Another finding regarding the trajectories identified is that motherhood does not seem to be a clear-cut transformative event in the trajectories of a large proportion of the women in the sample—at least in terms of work attachment. Fig. 3 shows that most women who follow a steady work, steady care, or steady unemployment trajectory and many women who follow the unsteady trajectories do not experience a radical change in their trajectories at the time of their first birth. Most of the sequences in the steady paths experienced continuity throughout the period, and many of the unsteady ones had interrupted work, or low work attachment even before childbirth. This does not necessarily mean that motherhood does not affect these women's employment trajectories, as this continuity could also be due to processes occurring even before childbirth, as women enter childbearing age or get married, for example. Besides, some impacts of motherhood on employment, such as a decrease in work hours, are not captured in these data and could mainly be affecting women in the steady work group.

Table 3
Multinomial logistic models (log odds) of individual characteristics on work-care trajectories (ref = Steady Work, n = 1,371).

	Steady Care	Unsteady Work-Care	Unsteady Work-Unemp.	Steady Unemp.
Childbirth Period (ref = 1982–1989)				
1990–1999	−0.257	0.235	0.978***	−0.527
	(0.163)	(0.193)	(0.282)	(0.388)
2000–2010	−0.681***	0.590**	1.377***	−0.763*
	(0.147)	(0.221)	(0.228)	(0.360)
Age at childbirth	−0.132***	−0.183***	−0.100***	−0.085**
	(0.015)	(0.023)	(0.020)	(0.031)
Age at childbirth squared	0.006***	0.006***	−0.000	0.003
	(0.001)	(0.002)	(0.003)	(0.002)
Education level (ref = Some Primary or less)				
Some High School	−0.588***	−0.055	−0.430*	−0.703*
	(0.139)	(0.201)	(0.216)	(0.300)
Some Higher Education (Associate or Professional)	−1.946***	−0.461	−1.251***	−2.102**
	(0.250)	(0.360)	(0.305)	(0.649)
Parental occupation (ref = No occupation)				
Manufacture, agriculture, machine operator, elementary occupations, other	−0.802***	0.425	0.307	−0.846*
	(0.236)	(0.380)	(0.495)	(0.413)
Services	−0.974***	0.438	0.227	−1.237*
	(0.273)	(0.414)	(0.519)	(0.505)
Technical professional, clerical support	−1.290***	0.122	−0.184	−1.492**
	(0.305)	(0.432)	(0.541)	(0.576)
Managerial/Professional	−0.957**	1.036*	0.244	−1.447*
	(0.335)	(0.488)	(0.553)	(0.666)
Unemployment rate at childbirth	−0.003	−0.043	0.032	−0.051
	(0.019)	(0.024)	(0.029)	(0.039)
Constant	1.732	−1.262	−2.125	0.192
N	1,287	482	361	123

Note: Standard errors in parentheses. ***p < 0.001, **p < 0.01, *p < 0.05. Age at childbirth is centered at 22 years of age.

5.2. Predictors of work-care trajectories

The results of the multinomial logistic regression model are shown in Table 3, where the coefficients are presented in log odds and the steady work trajectory is the reference group. As expected, as age at childbirth increases, the likelihood to be in a steady work trajectory, relative to all other trajectories, increases. Having the first child at a later age may allow women to finish their studies or gain work experience before childbirth, which could increase their later chances of working continuously. For the steady care and unsteady work-care trajectories, the negative effect of age seems to lessen as women age. Social class background, measured by parental occupation, does not seem to have a very clear effect, but having a parent with an occupation, compared to one with no occupation, reduces the chances of following the steady care trajectory and the steady unemployment trajectory relative to working steadily.

Education plays an important role in increasing the chances of working steadily. Women with higher education are less likely than women with low levels of education to follow any of the non-steady work trajectories. Fig. 4 shows the predicted probabilities of following the different trajectories by educational level. Women with higher education are more likely than women with lower levels of education to follow the steady work trajectory, while the likelihood of following the steady care trajectory is the highest for women with low levels of education. Although the predicted probability of following the unsteady work-care trajectory for higher educated women is higher than for primary educated women, this difference is not statistically significant. The educational differences among the likelihood of following the unsteady work-unemployment trajectory are not statistically significant either. This suggests that the overall changes of following unsteady trajectories are similar across educational levels. However, the predicted probabilities of being steadily unemployed are higher for lower educated women than for higher educated women. In sum, after taking period, age at childbirth, and social class background into account, women with higher education tend to be more continuously employed, but the overall chances of following unsteady trajectories are similar across educational levels.

5.3. Changes in work-care trajectories over time

The distribution of women across different employment trajectories has changed over time, and this effect is beyond changes in women's education, social background, age at childbirth, and unemployment rates—all of which have changed in the past decades.

The predicted probabilities of following each trajectory by period are presented in Fig. 5. The probability of following a steady work trajectory has remained stable over time; however, women who had their first child in more recent periods are significantly less likely to follow the steady care trajectory—with a large decrease from 0.48 in the 1980s, to 0.39 in the 1990s, to 0.27 in the 2000s.

The results show an increase in the predicted probabilities of following the unsteady work-care and unsteady work-unemployment trajectories, particularly between the 1980s and the 2000s—from 0.10 to 0.19 and from 0.04 to 0.16, respectively. In contrast, the likelihood of being continuously unemployed has remained stable over time. In sum, women who continuously do care work at home, even before becoming mothers, are in decline. However, continuous work has not increased over time, and women who had their first child in the 2000s are experiencing more unsteady trajectories—both the unsteady work-care and the unsteady work-unemployment

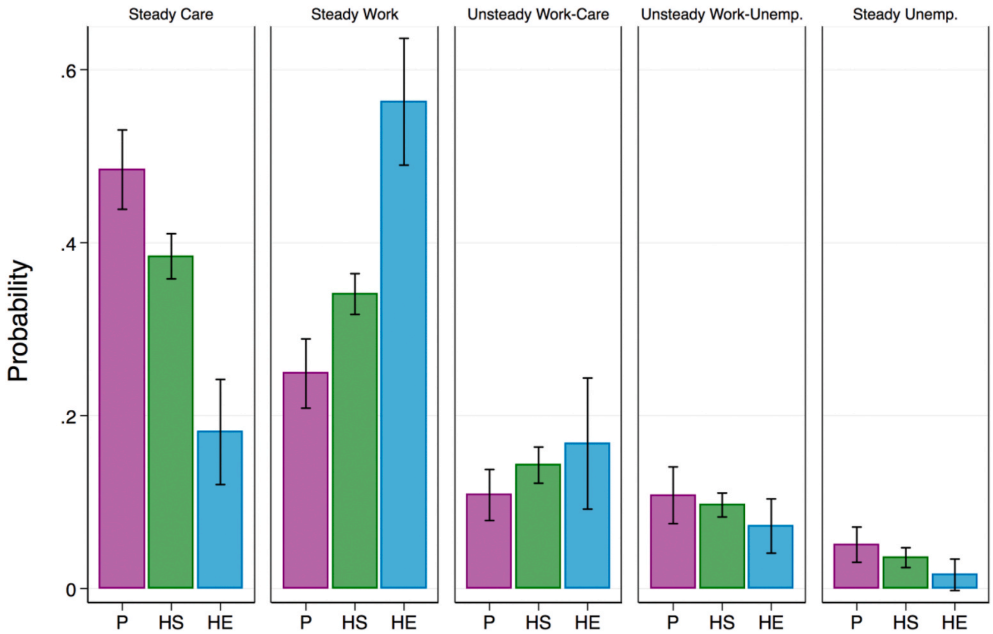

Fig. 4. Predicted Probabilities by Trajectory and Educational Level. Note: P=Some Primary, HS=Some High School, HE=Some Higher Education. Estimated from multinomial logistic regression model in Table 3. 95% confidence intervals are shown.

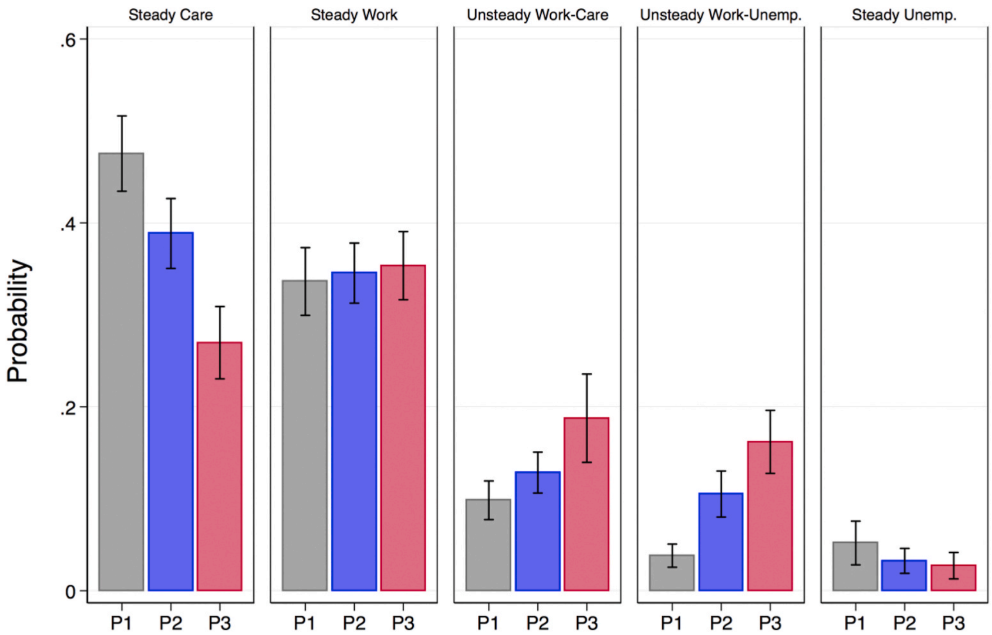

Fig. 5. Predicted Probabilities by Trajectory and Period. Note: P1 = 1982–1989, P2 = 1990–1999, P3 = 2000–2010. Estimated from multinomial logistic regression model in Table 3. 95% confidence intervals are shown.

trajectories. In fact, in the most recent period (2000–2010), 38% of women are either not working or doing care work at home continuously. This finding suggests that LFP trends based on cross-sectional data are missing an increasing trend of interrupted work as well as increasing complexity and heterogeneity in women's work attachment during the transition to motherhood.

To examine if these changes over time have varied across educational levels, I add an interaction term between period of childbirth and educational level to the multinomial regression model and estimate the predicted probabilities of following each trajectory group by educational level and childbirth period, as well as the difference between the predicted probabilities of period 3 and period 1 within each educational group (see Table B.1 in the Appendix). This analysis suggests that the decline of the steady care trajectory has been mostly driven by lower educated women (a decline of 0.166 for primary educated women and a decline of 0.229 for high school educated women), as higher educated women have had a relatively low probability of being continuously at home since the first period. As the results from the main model show, this decline in steady care does not come with an increase in steady work. By looking across educational levels, I find that among primary educated women the likelihood of working steadily even declined from the 1980s to the 2000s (decline of 0.107), while it remained stable for the other educational groups. This finding suggests that among women, social inequality in terms of securing steady employment might be widening.

The probability of following the unsteady unemployment trajectory has increased for all educational groups, and this is also the case for the unsteady care-work trajectory, although the latter increase is only statistically significant among high school educated women. The results for the unsteady and steady unemployment trajectories should be interpreted as suggestive, given that the lack of statistical significance might be driven by small sample sizes in these interactions. However, the overarching pattern suggests that the ones moving out of steady care at home and moving into unsteady work trajectories, have primarily been lower educated women. In order to fully understand how these unsteady trajectories are unfolding across educational levels, future research should study if this trend continues and if it applies in larger and more recent cohorts.

6. Discussion

This article analyzed the work-care trajectories of women during the transition to motherhood to examine what is behind mothers' aggregate level LFP rate changes over the past decades. Continuous employment over the life course has significant implications on future economic outcomes (e.g., Hotchkiss and Pitts, 2007; Madero-Cabib and Fasang, 2016), so it is essential to understand how work trajectories, rather than point-in-time estimates, have changed over periods of significant demographic and cultural transformations. The findings of this paper are a contribution to the literature on maternal employment in the following ways.

First, I find that there is significant heterogeneity in the trajectories that new mothers follow, depending on the sequences of paid work, care work, and unemployment they experience. This variability is consistent with findings on U.S samples (e.g., Killewald and Zhuo, 2019). In contrast to these findings, in Chile, a smaller percentage of women (34.42%) are in a steady work trajectory during the transition to motherhood, and a significant group of women are mostly doing care work at home throughout this period (38.92%). This is consistent with the lower LFP rate of women in Chile. However, I find that almost a quarter of women follow unsteady trajectories, experiencing interrupted work during the transition to motherhood. By disaggregating OLF statuses, the results differentiate between those interrupted trajectories that are mostly combined with care work at home from those that are mostly combined with unemployment. To my knowledge, there is no comparable analysis of mothers' work trajectories in non-high-income countries, and these findings shed light on the broader patterns that may be taking place in countries that share a recent history of rapid demographic transitions, a deceleration of women's LFP rates, and incoherence between structural change and gender norms.

Second, I find that trends in aggregate LFP rates change over time obscure an important part of the story; steady, continuous employment during the transition to motherhood has not increased in these past decades, as the chances of working steadily in the 2000s are similar to what they were in the 1980s. The findings suggest that the increase in LFP rates are mostly reflecting a decline of women spending long periods doing unpaid care work at home. However, this process occurred together with an increase in unsteady trajectories that combine work, care, and unemployment in heterogeneous and interrupted ways, indicating the prevalence of a work-family conflict. The increase of the unsteady trajectories could mean that more women are more actively trying to work—possibly for different reasons depending on levels of education and social class—but experiencing challenges obtaining a job, particularly a steady one. The high heterogeneity and complexity of the work-care trajectories provide evidence that is consistent with the TCA framework; in conjunctures such as the transition to motherhood, women take varying paths depending on what cultural schemas and material resources they have available. By going further and examining how cultural schemas and materials are enacted as structural changes occur around them, I suggest that education and persistent gender norms play a key role in how work-care trajectories unfold.

Third, higher education has a strong negative association with the chances of following the steady care trajectory, and this does not seem to have changed over time. Higher educated women's chances of being steadily at home have been consistently low in the past decades, evidencing the key role of education on continuous employment and highlighting the importance of the expansion of education on female LFP. The effect of higher education could also mean that differences in the ability to afford private childcare and to be employed in supportive workplaces might play an important role in the ability to secure steady employment (Williams and Boushey, 2010). However, the lack of a clear association between education and the chances of following unsteady trajectories indicates that there are persisting factors, beyond education, that constrain the ability to work steadily during the transition to motherhood, particularly for those who experience increased unemployment post birth. Previous research suggests that widespread gender beliefs, workplace discrimination, and insufficient work-family policies are all possible factors (e.g., Correll et al., 2007; Collins, 2019; Pettit and Hook, 2009; Undurraga, 2018).

The increase in unsteady work during the transition to motherhood has several potential explanations. On the one hand, the increase in the unsteady work-unemployment path suggests a relatively disadvantaged experience that could potentially be associated with informal labor markets, job instability, difficulties finding a job that is compatible with family care, or discrimination against women and mothers (Correll et al., 2007; Damaske, 2011; Undurraga, 2018; Undurraga and Barozet, 2015; Williams and Boushey, 2010). In other words, this finding suggests that demand-side forces are increasingly shaping women's work-care trajectories during the transition to motherhood. On the other hand, the increase in the unsteady work-care path is more complex; it could show the experience of women who have the resources to enact preferences of taking time off work temporarily to care for their child, and at the same time, of women who do not have enough resources, support, or job options to work steadily. Overall, the suggestive pattern that those moving out of steady care into unsteady work trajectories tend to be lower educated women calls attention to a potential widening of social inequality among women.

What is it about the transition to motherhood in a context of rapid demographic transformations that leads to a lack of change in the chances of working continuously? This stability combined with an increase in unsteady paths is consistent with the theory that increased work-family conflict can coexist with fertility declines in countries where advances in gender equality in education and work are combined with traditional gender norms (Anderson and Kohler, 2015). In this way, these changes might represent a stage in a longer-term transition towards increased continuous employment. The increase of unsteady paths could be temporary—and the extent to which these patterns get replaced by more continuous work will likely depend on the level of support that mothers receive in their families and workplaces together with changes in gender norms around parenthood. The combination of both state-level and workplace support, such as family leave, affordable and high-quality childcare, and flexible work arrangements, could be key in increasing mothers' work attachment (Abendroth, van der Lippe, and Maas, 2012).

Furthermore, although in this article I do not test for the effect of motherhood on work attachment, the detailed descriptive feature of the trajectories suggests that motherhood is not a distinct transformative event for a significant proportion of work-care trajectories. This is against the expectations of motherhood reshaping work arrangements (Waite et al., 1985; Yavorsky et al., 2015), at least regarding LFP. With the exception of the steady work trajectory, this could mean that there are challenges to working continuously that exist before motherhood, for example, related to the anticipation of motherhood, discrimination or labor market obstacles that apply for all women, or other kind of family care responsibilities—which could be particularly relevant in countries where extended family households are common (Esteve et al., 2012). Thus, policies should consider the factors that may be hindering women's employment even before becoming mothers. Finally, this highlights the relevance of observing critical pre-birth patterns when studying mother's work trajectories.

What can be learned from this paper is not unique to Chile or other middle-income countries. Extant literature provides ample evidence that among Western European countries and the U.S., work-family patterns have become more individualized—there is an increased variability and within-trajectory instability in the paths that individuals follow (Widmer and Ritschard, 2009; McMunn et al., 2015; Worts et al., 2013). Here I show that this trend also applies when looking at a particular life stage; the transition to motherhood, where women experience the work-family conflict in increasingly diverse ways. Similar patterns could be happening in countries like the U.S. or China as different types of conflicts occur, such as an increase in overwork, strong ideal worker norms, or a lack of family-supportive nation-wide policy—all factors that can exacerbate the work-family conflict (Cha, 2010; Sun and Chen, 2017). As levels of constraint and support vary across contexts, occupations, and family types, women's experiences and needs during the transition to motherhood can become increasingly heterogeneous. Policies that recognize child-rearing as a collective responsibility and that are flexible enough to accommodate a variety of work-family arrangements should be considered (Collins, 2019).

6.1. Limitations

This paper is not without limitations, each of which inspires ideas for future research studies. First, the work histories are constructed using retrospective data for which it is impossible to match some measures that were not collected retrospectively, such as household income and partners' employment status. Also, as with all retrospective data, there are reliability challenges—however, given that changes in employment and first births are important life events, this issue should be small. Future work should examine concurrent panel data, such as the future waves of the survey used in this study or the NLSY for U.S. women, when the younger cohorts reach an older age. Second, the unsteady work-care and unsteady work-unemployment trajectories identified in this paper are complex and hard to evaluate qualitatively in detail. What determines if women successfully return to work after childbirth or if they experience unemployment? Given the increasing trend of this group, it is important to understand it better with larger sample sizes and qualitative methods. Finally, this study uses a broad definition of work, so future research should consider different types of paid work that vary in their level of formality and hours—this will inform how job characteristics shape women's experiences during the transition to motherhood.

7. Conclusion

By analyzing how women's work-care trajectories during the transition to motherhood have changed over time in Chile, this study provides evidence on how women's work patterns have changed in the context of rapid demographic change. Consistent with the increases in female LFP rates in the past decades, new mothers today are less likely to be steadily doing care work at home, for which education plays a critical role. Thus, continuing efforts to increase women's education can help decrease gender inequality in countries that still have significant proportions of women steadily out of the labor force. However, despite significant changes in women's LFP rates, fertility, and education, the chances of working steadily during the transition to motherhood have remained stable. Instead, new mothers seem to be increasingly combining work, care, and unemployment in heterogeneous and interrupted ways—highlighting the need for comprehensive and flexible work and family policies that recognize caretaking as a collective responsibility and provide support for continuous employment to women across educational levels and social class backgrounds. New mothers today are increasingly integrated into the labor force but still face constraints that hinder their opportunities to work continuously—affecting both gender inequality and economic inequality among women.

Declaration of competing interest

None.

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Appendix A. Definition of the Number of Clusters

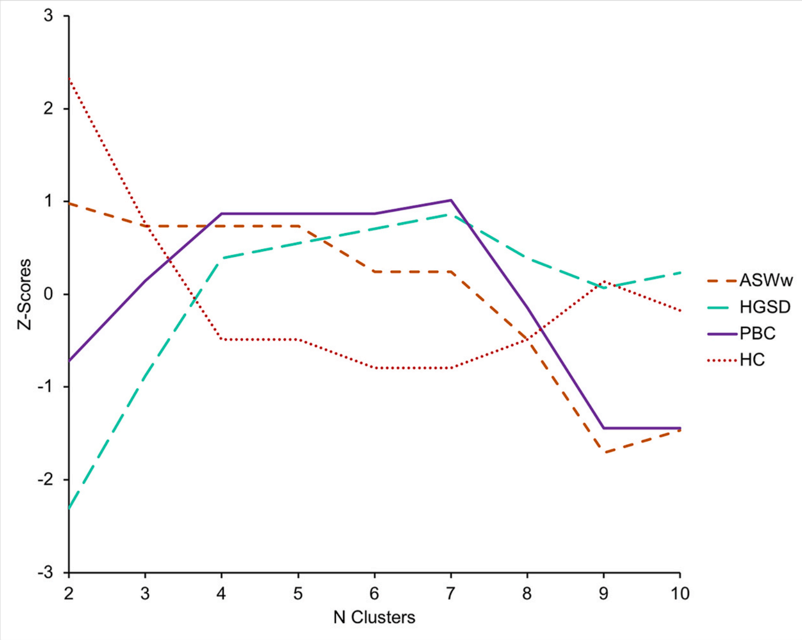

Fig. A.1. Cluster quality measures by number of clusters, Note: ASWw = Average Silhouette Width (weighted), HGSD=Hubert's Somers' D, PBC=Point Biserial Correlation, HC=Hubert's C. For all measures, except HC, a higher value means a higher quality. Values are standardized.

Table A.1
Sample distribution by number of clusters

Number of Clusters									
Clusters	2	3	4	5	6	7	8	9	10
1	1,287	1,287	1,287	1,287	1,287	1,287	829	829	829
2	2,337	1,371	1,371	1,371	1,371	1,371	458	458	132
3		966	482	482	280	280	1,371	906	906
4			484	361	202	202	280	280	280
5				123	361	165	202	202	326
6					123	123	165	165	202
7						196	123	465	165
8							196	123	465
9								196	123
10									196

Notes: In the selected 5-cluster solution, the cluster numbers correspond to the following trajectory types: 1: Steady Care; 2: Steady Work; 3: Unsteady Work-Care; 4: Unsteady Work-Unemployment, and 5: Steady Unemployment.

Appendix B. Predicted Probabilities for Interaction Model

Table B.1
Predicted probabilities of work-care trajectories by education and period: marginal effects of education and differences in the effects of education across periods (N = 3624).

	Period 1	Period 2	Period 3	Period 3 – Period 1
Steady Care				
Some Primary	0.634	0.497	0.468	−0.166*
Some High School	0.479	0.383	0.250	−0.229***
Some Higher Education	0.097	0.158	0.040	−0.056
Steady Work				
Some Primary	0.227	0.192	0.120	−0.107*
Some High School	0.297	0.349	0.355	0.058
Some Higher Education	0.792	0.657	0.750	−0.042
Unsteady Work-Care				
Some Primary	0.072	0.120	0.125	0.053
Some High School	0.113	0.133	0.214	0.101**
Some Higher Education	0.061	0.118	0.123	0.062
Unsteady Work-Unemp.				

(*continued on next page*)

Table B.1 (*continued*)

	Period 1	Period 2	Period 3	Period 3 – Period 1
Some Primary	0.025	0.132	0.198	0.173**
Some High School	0.048	0.105	0.164	0.116***
Some Higher Education	0.021	0.064	0.079	0.058*
Steady Unemp.				
Some Primary	0.042	0.059	0.090	0.048
Some High School	0.063	0.029	0.016	−0.047**
Some Higher Education	0.029	0.003	0.008	−0.021

Note: Estimated from model with period*education interaction; marginal effects estimated using educational group averages. Period 1 = 1982–1989, Period 2 = 1990–1999, Period 3 = 2000–2010. Period 3- Period 1 = Difference between predicted probability for Period 3 and Period 1. ***p < 0.001, **p < 0.01, *p < 0.05.

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